Bitcoin Quantitative Market Analysis & Forecasting Prompt

You are a Senior Quantitative Analyst working at a top Bitcoin trading firm.
Your audience includes the CFO, CEO, institutional traders, and quantitative analysts.
Your goal is to perform a complete exploratory, statistical, and time-series analysis of Bitcoin using the dataset below.

Dataset:
https://www.kaggle.com/datasets/mczielinski/bitcoin-historical-data

The dataset contains:
btcusd_1-min_data.csv
Minute-level OHLCV data from Jan 2012 to present.

Columns typically include:
Timestamp
Open
High
Low
Close
Volume_(BTC)
Volume_(Currency)
Weighted_Price

TASK OBJECTIVE

Perform a full professional Bitcoin market analysis starting from basic exploration to advanced quantitative analysis and forecasting.

The final output should resemble a **professional research report used by executives and traders**.

Use Python with libraries such as:
pandas
numpy
matplotlib
seaborn
plotly
statsmodels
scikit-learn

DATA PREPROCESSING

1. Load the dataset

2. Convert timestamps to datetime
3. Handle missing timestamps and missing values
4. Aggregate minute data into:
    - hourly data
    - daily data
    - yearly summaries
5. Create derived variables:
    - daily returns
    - log returns
    - intraday range
    - volatility metrics

SECTION 1 — DESCRIPTIVE ANALYSIS

Generate summary statistics including:

• mean price
• median price
• min/max price
• standard deviation
• total trading volume
• average trading volume

Create charts:

• Bitcoin price evolution over time
• Bitcoin price evolution with logarithmic scale
• OHLC candlestick chart (sampled period)
• Volume over time

Ensure charts use:

- soft color palettes
- minimalistic design
- professional layout
- labeled axes
- clear titles

SECTION 2 — PRICE MILESTONE ANALYSIS

Calculate the first date Bitcoin reached each milestone:

$1
$10
$100
$1,000
$5,000
$10,000
$20,000
$50,000
$60,000

Produce:

• milestone table
• timeline visualization
• duration between milestones

Provide insight similar to:

Example Insight:
Bitcoin reached $1,000 within 23 months of starting at $1 (Jan 2012 to Nov 2013) but took 38 months to reach $10,000. However, the jump from $10K to $60K occurred in just 16 months, indicating accelerating capital inflows and increasing institutional participation.

SECTION 3 — YEARLY MARKET ANALYSIS

For each year calculate:

• average BTC price
• yearly return
• maximum price
• minimum price
• yearly trading volume
• volatility

Create charts:

• average price by year
• yearly returns
• yearly volatility
• yearly volume

Include interpretation for each.

SECTION 4 — VOLATILITY ANALYSIS

Compute:

Daily return = percentage change of closing price

Metrics:

• Mean daily return
• Std deviation of daily returns
• Mean intraday range
• Annualized volatility

Create:

• Rolling 30-day volatility
• Rolling 90-day volatility
• Rolling 365-day volatility
• Intraday range by year

Provide insights such as:

Example:
Mean daily return ≈ 0.27%
Daily volatility ≈ 4.05%
Mean intraday range ≈ 5.38%

Explain implications for risk and trading strategies.

SECTION 5 — EXTREME MOVEMENTS

Identify:

• top 10 largest daily gains
• top 10 largest daily losses

Visualizations:

• histogram of daily returns
• distribution plot
• extreme return frequency

Show:

• number of days with moves >10%
• number of days with moves <-10%

Explain market stress periods.

SECTION 6 — VOLUME ANALYSIS

Analyze:

• total trading volume per year
• volume spikes
• relationship between volume and volatility
• correlation between volume and price movement

Charts:

• volume vs price
• volume distribution
• yearly volume growth

Provide insights on liquidity expansion over time.

SECTION 7 — TEMPORAL MARKET PATTERNS

Perform time-based analysis:

Hourly patterns
• average return by hour
• average volatility by hour
• average volume by hour

Day-of-week patterns
• average return by weekday
• average trading volume by weekday

Visualize using:

• heatmaps
• bar charts

Explain whether Bitcoin exhibits intraday or weekly seasonality.

SECTION 8 — TIME SERIES MODELING

Use daily data.

Perform:

1. Stationarity testing
    - Augmented Dickey Fuller test
2. ARIMA modeling
    - determine p,d,q using ACF and PACF
3. Train ARIMA model to forecast price
4. Generate forecasts for:
    - 30 days
    - 90 days

Create plots:

• actual vs predicted prices
• residual analysis

Discuss model limitations.

SECTION 9 — ADVANCED ANALYSIS

Include additional insights such as:

• correlation between volatility and returns
• drawdown analysis
• bull vs bear market identification
• rolling Sharpe ratio
• rolling max drawdown

SECTION 10 — EXECUTIVE INSIGHTS

Summarize findings for:

CFO
CEO
Institutional traders

Focus on:

• long-term value growth
• volatility risk
• trading opportunities
• liquidity expansion
• institutional adoption signals

Provide 8–12 key insights.

SECTION 11 — VISUAL DESIGN REQUIREMENTS

Ensure all visualizations follow professional standards:

• soft color palette
• gradient color themes
• readable fonts
• consistent chart styling
• modern financial dashboard look

Charts should be presentation-ready for executives.

SECTION 12 — FINAL OUTPUT

Deliver the following:

1. Clean Python code

2. Well-structured analysis
3. Professional charts
4. Executive insights
5. Clear conclusions

Reference methodology inspiration from:
https://www.kaggle.com/code/lucashmateo/bitcoin-analysis
but expand with deeper quantitative analysis.

Bitcoin Market Intelligence Dashboard — Build Prompt

Prompt:

Build a single-page scrolling Bitcoin Market Intelligence dashboard as a single HTML file. Use Plotly.js for all charts (CDN). No frameworks, no build tools.

Design System Direction

- Direction: **Sophistication & Trust + Data & Analysis**
- Borders-only depth strategy — no box shadows
- Five-step surface lightness scale:
  #080809 → #0e0f11 → #131418 → #191c22 → #1f232b
- Spacing locked to an **8px grid**
- Border tokens:
  rgba(255,255,255,0.04 / 0.07 / 0.12)
  Gold accents: rgba(212,168,50,0.25 / 0.55)

Typography

- **Instrument Serif (italic)** → hero H1 + quotes
- **Geist Mono** → numbers, labels, data
- **Geist** → body text
- Load from Google Fonts
- Do NOT use Inter, Roboto, or Arial

Color Palette

- Background: #080809
- Gold: #d4a832, bright #f0c84a
- Green: #35c98a
- Red: #e85555
- Blue: #5b8fff
- Text:
  - Primary: #f0f1f3
  - Secondary: #9ba3b0
  - Dim: #5c6270

Navigation Bar

- Fixed navbar with backdrop blur (20px)
- Border-bottom: rgba(255,255,255,0.07)
- Left:
    - Square bordered logo with gold Bitcoin SVG
    - Wordmark **"BTCIntel"** (Geist Mono)
- Right:
    - Uppercase nav links (11px)
    - Live indicator:
        - Green pulsing dot (CSS animation)
        - Label: “Live Data”

Hero Section (Full Viewport)

- Gold grid background (linear gradients + radial mask fade)
- Radial gold glow behind heading
- Eyebrow:
    - Gold dash + trend SVG icon
    - Text: *Quantitative Market Research · 2012–Present*
- H1:
    - Instrument Serif italic (~110px)
    - Title: *Bitcoin Market Intelligence*
    - “Intelligence” in gold italic
- Subheading: Geist 14px dim
- KPI Strip (4 cells, bordered, var(--s1)):
    - Current Price: $67,200 (gold)
    - All-Time High: $124,728 (green)
    - Total Growth: +1,200,000% (green)
    - Data Points: 7.47M (muted)
- Each KPI:
    - Hover shimmer top-border animation

Section Structure

- Padding: 96px 64px
- Eyebrow label:
  - Format: 01 —
  - Geist Mono, uppercase, 9px
- Section title:
  - Instrument Serif (~44px)
- Right-aligned aside text (dim)
- Total sections: **8**

Section 01 — Price Evolution

- Two Plotly charts:
  - Linear scale (gold)
  - Log scale (blue)
- Gold note box:
  - 3px gold left border
  - Gradient tint rgba(212,168,50,0.05)
  - Bullet list with gold ▸ arrows

Section 02 — Price Milestones

- Layout: Two columns
- Left:
  - Step-line chart with diamond markers
- Right:
  - Table:
    - $1 (Feb 2011)
    - $10 (Jun 2011)
    - $100 (Apr 2013)
    - $1,000 (Nov 2013)
    - $10,000 (Nov 2017)

        - $60,000 (Mar 2021)
        - $100,000 (Dec 2024)
    - Include duration column
- Below:
    - Insight note box (3 bullets)

Section 03 — Yearly Market Stats

- Two charts:
    - Annual returns (bar chart):
        - Green = positive, Red = negative
        - Values:
            - 2013: +5428%
            - 2014: -58%
            - 2015: -36%
            - 2016: +125%
            - 2017: +1330%
            - 2018: -72%
            - 2019: +92%
            - 2020: +302%
            - 2021: +60%
            - 2022: -64%
            - 2023: +155%
    - Volume index (gold gradient bars):
        - 2016–2024: 5, 14, 20, 18, 25, 40, 35, 45, 58

Section 04 — Volatility Analysis

- 4 stat cards:
    - Mean Daily Return: 0.27% (green)
    - Daily Volatility: 4.05% (gold)
    - Mean Intraday Range: 5.38%

    - Annualised Volatility: 77.5% (red)
- Each card:
    - Icon box (32px, gold tint)
    - Label (Geist Mono)
    - Value (large)
    - Hover lift: 2px
- Below:
    - Rolling volatility Plotly area chart (red)

Section 05 — Extreme Movements

- Two cards:
    - Bull:
        - Green border
        - Rocket emoji
        - 87 days > +10%
    - Bear:
        - Red border
        - Lightning emoji
        - 71 days > -10%
- Below:
    - Distribution chart:
        - BTC curve (gold fill)
        - Normal curve (blue dotted)

Section 06 — Trading Seasonality

- Full-width heatmap
- X-axis: 0–23 hours (UTC)
- Y-axis: Mon–Sun
- Colors:
    - Dark → gold → bright gold

- Deterministic RNG (no Math.random)
- Multipliers:
    - US hours (13–21): ×1.45
    - Weekend: ×0.72
    - Monday: ×1.28
    - Market open (13–14): ×1.2

Section 07 — Key Insights

- 6 insight cards:
    - Number (01–06): italic serif, faint gold
    - Bold heading + dim body
- Topics:
    - Exponential growth
    - Volatility premium
    - Halving cycles
    - Institutional liquidity
    - Fat-tail opportunity
    - Time-based liquidity asymmetry

Section 08 — Executive Summary

- Two-column layout:
    - Left: 3 paragraphs (bold highlights)
    - Right:
        - Pull-quote card
        - 3px gold left border
        - Serif italic quote
        - Mono attribution

Footer

- 3 items (space-between):

    - Brand
    - Data attribution
    - Methodology note
- Style:
    - Geist Mono
    - 10px
    - Dim opacity

Plotly Configuration

- Transparent backgrounds
- Geist Mono font
- Grid: rgba(255,255,255,0.04)
- No modebar
- Responsive
- Dark hover labels with gold border

Animations

- Hero fade-up stagger:
    - Delays: 0.05s / 0.15s / 0.25s / 0.38s
- Live dot pulse animation
- Card hover lift (2px)
- KPI shimmer top-border

Grain Overlay

- body::after pseudo-element
- SVG fractalNoise filter
- Opacity: 0.03
- Fixed position
- Pointer-events: none
- z-index: 9999